Music Bot: Step-by-Step Feature Guide

1. Get started

1. Install the dependencies in **requirements.txt** and run **python main.py** on a persistent worker with FFmpeg.
2. Configure **API_ID**, **API_HASH**, **BOT_TOKEN** (or **TELEGRAM_BOT_TOKEN**), and **SESSION_STRING**. Set **OWNER_ID** to your numeric Telegram user ID to enable owner maintenance commands. Keep these values private.
3. Add the bot and its assistant account to your Telegram group and start a voice chat. The assistant must be able to join it.
4. Open the bot privately and send **/start**, or send **/start** in the group to see the command summary. Telegram's command menu lists the main features.

2. Play music and video

- **/play Believer** searches for music and starts playback, or adds it to the queue if something is playing. **/song** is an alias. A supported URL also works.
- Reply to an audio file with **/play** to play it.
- **/vplay <name or URL>** searches for and plays video in the voice chat. Reply to a Telegram video with **/vplay** to play that file. Video needs a group call that supports video and sufficient bandwidth.
- **/search <name>** shows five choices. The requester must select a result within five minutes.

3. Control playback

1. Send **/pause** to pause and **/resume** to continue.
2. Send **/skip** to move to the next item; **/next** does the same. **/voteskip** lets members vote to skip.
3. Send **/replay** to start the current item again. This replaces the former **/restart** track command.
4. Send **/stop** or **/end** to leave the call, clear the queue, and cancel pending requests.
5. Send **/seek 1:30** or **/seek 90** to jump to a position before the end. The duration must be known.
6. Send **/volume 80** to set the volume from 1 to 200. **/loop** toggles repeating the queue; **/shuffle** mixes upcoming items only.
7. Use the buttons on the current song card for common controls. Old cards cannot control a new song.

4. Manage the queue and playlists

- **/queue** displays the current item and numbered upcoming items. **/remove 2** removes upcoming item 2. **/clearqueue** removes upcoming items and cancels pending downloads while the current item continues.
- **/playlist save favourites** saves the current item. **/playlist list** lists playlists; **/playlist show favourites** displays its items.
- **/playlist play favourites** loads its items. **/playlist remove favourites 2** removes an item; **/playlist delete favourites** deletes the playlist. Playlists are shared within the group and can hold up to 100 items each.

5. Group settings and permissions

1. Send **/settings** to see the group control mode, autoplay setting, and skip vote threshold.
2. A group admin can send **/mode admin** to restrict playback controls to group admins and authorized users, or **/mode all** to allow all members. **/admincontrols on** and **/admincontrols off** are equivalent. Song requests, playlists and skip voting remain available to members.
3. To authorize a user, an admin replies to that user's message with **/auth**, or sends **/auth 123456789** with their numeric user ID. Use **/unauth** in the same way to remove authorization. Authorization applies to that group only.
4. An admin can send **/autoplay** to toggle it, or **/autoplay on** or **/autoplay off**. When the queue finishes naturally, the bot searches using the last title and tries to play a different result. Looping the queue takes precedence. Autoplay may fail when no suitable result is found or a source is unavailable.
5. An admin can send **/votethreshold 3** to require three distinct skip votes; choose a number from 1 to 20. Changing the threshold clears existing votes.

6. Maintenance

- **/speed** measures the bot's response time to a Telegram message, in milliseconds.
- The configured owner can use **/clean** to remove expired in-memory search results. Temporary downloaded media is removed as items finish, are skipped, or are stopped.
- The configured owner can use **/reboot**, **/reload**, or **/restart** to restart the bot worker and reload its code. This clears active queues and reloads the built-in handlers. There is no separate plugin system or host machine restart in this project. Deploy changed files before reloading.
- The bot must be restarted after changing environment variables. Host process managers and environment settings control deployment and machine restarts.

7. Data and troubleshooting

- Group settings, authorized users and playlists use SQLite. Set **MUSIC_DB_PATH** to a writable file on persistent storage, such as **/data/music.sqlite3**, and back it up. Queues, votes and temporary search results are in memory and reset on restart.
- If playback fails, check that the group voice chat is active, the assistant account can join, FFmpeg is available, and the media source is accessible. Try another source if a download fails.
- The bot checks admin status through Telegram. Anonymous admins should send settings commands from their user account. Owner commands require a numeric **OWNER_ID** matching the sender.
- Use media you have permission to stream. Test the bot in a real group after deployment.